

The pkginfograb Package

Version 1.2

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Abstract

This package is aimed at package writers and offers a way to collect/document L^AT_EX package's info (name, version, description, etc.) in a systematic way, including a mechanism to check package's version. Just a few functions are defined, to document/set package's info, retrieve them and to verify package's version (if it is newer than a given reference).

1 Expl3 Commands

`\pkginfograb_set:nn` `\pkginfograb_set:nn` {<pack-name>} {<keyval-list>}

This will create a property list associated with <pack-name>. <keyval-list> might contain any set of keys, though, the functions below expect at least <version> or <date> (for version checking) and <name>, <date>, <version> and <description> (for `\pkginfograb_description:n`).

Note: An error will be raised if calling it twice for the same <pack-name>.

For Example:

```
\pkginfograb_set:nn {pkginfograb}
{
  name      = {pkginfograb} ,
  prefix    = {pkginfograb} ,
  date      = {2025/11/16},
  version   = {1.2} ,
  description = {Collecting~ package's~ info~ in~ a~ regular~ way}
}
```

`\pkginfograbProvidesExplPackage` `\pkginfograbProvidesExplPackage` {<pack-name>} {<keyval-list>}

Same as `\pkginfograb_set:nn`, but calling `\ProvidesExplPackage` right after, with the information just set. It assumes the following properties to be present: *name*, *date*, *version* and *description*. This is meant to de-duplicate information (and reduce the chances of inconsistencies). This doesn't follow *expl3* convention, since *expl3* code régime will starts only after `\ProvidesExplPackage`.

For Example:

```
\pkginfograbProvidesExplPackage {newpackage}
{
  name      = {newpackage} ,
  prefix    = {newpack} ,
  date      = {2025/11/16},
  version   = {1.0a} ,
  description = {some~ lazy~ pack}
}
```

This will be equivalent to call `\ProvidesExplPackage {newpackage} {2025/11/16} {1.0a} {some~ lazy~ pack}` after `\pkginfograb_set:nn`.

Note: *expl3* code régime will be en force after it.

*<https://github.com/alceu-frigeri/pkginfograb>

`\pkginfograb_req_version:nnn` `\pkginfograb_req_version:nnn` $\{\langle\text{your-pack}\rangle\}\{\langle\text{pack-name}\rangle\}\{\langle\text{min-version}\rangle\}$

This will verify if $\langle\text{pack-name}\rangle$'s $\langle\text{version}\rangle$ (as stored with `\pkginfograb_set:nn`) is at least $\langle\text{min-version}\rangle$. It expects $\langle\text{version}\rangle$ in one of three formats $[v]\text{digits}[\text{letters}]$, $[v]\text{digits.digits}[\text{letters}]$ or $[v]\text{digits.digits.digits}[\text{letters}]$ (the $[v]$, if present, is ignored).

Note: An error will be raised if $\langle\text{pack-name}\rangle$'s info isn't defined, incorrect version format or $\langle\text{min-version}\rangle$ isn't satisfied, in which case the error will note that **your-pack** needs version $\langle\text{min-version}\rangle$ of $\langle\text{pack-name}\rangle$.

`\pkginfograb_req_date:nnn` `\pkginfograb_req_date:nnn` $\{\langle\text{your-pack}\rangle\}\{\langle\text{pack-name}\rangle\}\{\langle\text{min-date}\rangle\}$

This will verify if $\langle\text{pack-name}\rangle$'s $\langle\text{date}\rangle$ (as stored with `\pkginfograb_set:nn`) is at least $\langle\text{min-date}\rangle$. It expects $\langle\text{date}\rangle$ in one of three formats $YYYY/MM/DD$, $YYYY-MM-DD$ or $YYYY.MM.DD$.

Note: An error will be raised if $\langle\text{pack-name}\rangle$'s info isn't defined, incorrect version format or $\langle\text{min-date}\rangle$ isn't satisfied, in which case the error will note that **your-pack** needs version $\langle\text{min-date}\rangle$ of $\langle\text{pack-name}\rangle$.

`\pkginfograb_set_aliases:` `\pkginfograb_set_aliases:`

This will set L^AT_EX 2_ε aliases for the `expl3` in this package. Note that none of the commands in 2 are defined by default, except `\PkgInfoSetAliases` which is an alias for this command.

`\pkginfograb_get:nn` ★ `\pkginfograb_get:nn` $\{\langle\text{pack-name}\rangle\}\{\langle\text{key}\rangle\}$

This will retrieve $\langle\text{key}\rangle$'s value. If $\langle\text{pack-name}\rangle$ or $\langle\text{key}\rangle$ doesn't exist, this will expand to nothing.

`\pkginfograb_get:nnN` `\pkginfograb_get:nnN` $\{\langle\text{pack-name}\rangle\}\{\langle\text{key}\rangle\}\{\langle\text{tl-var}\rangle\}$

This will store $\langle\text{key}\rangle$'s value at $\langle\text{tl-var}\rangle$. If $\langle\text{pack-name}\rangle$ or $\langle\text{key}\rangle$ doesn't exist, $\langle\text{tl-var}\rangle$ will be cleared.

`\pkginfograb_description:n` `\pkginfograb_description:n` $\{\langle\text{pack-name}\rangle\}$

This will typeset a small paragraph (for data validation) with following $\langle\text{pack-name}\rangle$'s info: $\langle\text{name}\rangle$, $\langle\text{version}\rangle$, $\langle\text{date}\rangle$ and $\langle\text{description}\rangle$

For example:

<code>\ExplSyntaxOn</code> <code>\pkginfograb_description:n{pkginfograb}</code> <code>\ExplSyntaxOff</code>	Package pkginfograb Version: 1.2 - 2025/11/16 <i>Collecting package's info in a regular way</i>
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`\pkginfograb_if_set_p:n` ★ `\pkginfograb_if_set_p:n` $\{\langle\text{pack-name}\rangle\}$

`\pkginfograb_if_set:nTF` ★ `\pkginfograb_if_set:nTF` $\{\langle\text{pack-name}\rangle\}\{\langle\text{if-true}\rangle\}\{\langle\text{if-false}\rangle\}$

new: 2025/11/01

This will test if the given $\langle\text{pack-name}\rangle$ was set with `\pkginfograb_set:nn` and $\langle\text{if-true}\rangle$ or $\langle\text{if-false}\rangle$ will be properly executed.

`\pkginfograb_map_inline:n` `\pkginfograb_map_inline:n` $\{\langle\text{inline-code}\rangle\}$

new: 2025/11/01

$\langle\text{inline-code}\rangle$ will receive (as $\langle\#1\rangle$) the name of each package set with `\pkginfograb_set:nn` in the order they were set.

<code>\ExplSyntaxOn</code> <code>\pkginfograb_map_inline:n</code> <code>{</code> <code> \pkginfograb_description:n{#1}</code> <code>}</code> <code>\ExplSyntaxOff</code>	Package pkginfograb Version: 1.2 - 2025/11/16 <i>Collecting package's info in a regular way</i> Package codedescribe Version: 1.16b - 2025/10/30 <i>L^AT_EX Code Description/Documentation</i> Package xpeekahead Version: 1.3a - 2025/11/01 <i>A simple peek ahead set up</i> Package codelisting Version: 1.16b - 2025/10/30 <i>L^AT_EX Code Listing</i>
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2 LaTeX2e Commands' Aliases

All commands below are aliases to their `expl3` counterparts. Aside from `\PkgInfoSetAliases` all other LaTeX2e aliases aren't defined by default. Call either `\pkginfograb_set_aliases:` or `\PkgInfoSetAliases` before using them.

`\PkgInfoSet` `\PkgInfoSet {⟨pack-name⟩}{⟨keyval-list⟩}`

This will create a property list associated with `⟨pack-name⟩`. `⟨keyval-list⟩` might contain any set of keys, though, the functions below expect at least `⟨version⟩` (for version checking) and `⟨name⟩`, `⟨version⟩`, `⟨date⟩` and `⟨description⟩` (for `\PkgInfoDescription`).

Note: An error will be raised if calling it twice for the same `⟨pack-name⟩`.

For Example:

```
\PkgInfoSet {pkginfograb}
{
  name      = {pkginfograb} ,
  prefix    = {pkginfograb} ,
  date      = {2025/11/16},
  version   = {1.2} ,
  description = {Collecting~ package's~ info~ in~ a~ regular~ way}
}
```

`\PkgInfoReqVersion` `\PkgInfoReqVersion {⟨your-pack⟩}{⟨pack-name⟩}{⟨min-version⟩}`

This will verify if `⟨pack-name⟩`'s `⟨version⟩` (as stored with `\PkgInfoSet`) is at least `⟨min-version⟩`. It expects `⟨version⟩` in one of three formats `[v]digits[letters]`, `[v]digits.digits[letters]` or `[v]digits.digits.digits[letters]` (the `[v]`, if present, is ignored).

Note: An error will be raised if `⟨pack-name⟩`'s info isn't defined, incorrect version format or `⟨min-version⟩` isn't satisfied, in which case the error will note that `your-pack` needs version `⟨min-version⟩` of `⟨pack-name⟩`.

`\PkgInfoReqDate` `\PkgInfoReqDate {⟨your-pack⟩}{⟨pack-name⟩}{⟨min-date⟩}`

This will verify if `⟨pack-name⟩`'s `⟨date⟩` (as stored with `\PkgInfoSet`) is at least `⟨min-date⟩`. It expects `⟨date⟩` in one of three formats `YYYY/MM/DD`, `YYYY-MM-DD` or `YYYY.MM.DD`.

Note: An error will be raised if `⟨pack-name⟩`'s info isn't defined, incorrect version format or `⟨min-version⟩` isn't satisfied, in which case the error will note that `your-pack` needs version `⟨min-date⟩` of `⟨pack-name⟩`.

`\PkgInfoSetAliases` `\PkgInfoSetAliases`

This will set LaTeX_{2 ϵ} aliases for the `expl3` in this package. Note that none of the commands in 2 are defined by default, except this.

`\PkgInfo` ★ `\PkgInfo {⟨pack-name⟩}{⟨key⟩}`

This will retrieve `⟨key⟩`'s value. If `⟨pack-name⟩` or `⟨key⟩` doesn't exist, this will expand to nothing.

`\PkgInfoGet` `\PkgInfoGet {⟨pack-name⟩}{⟨key⟩}{⟨macro⟩}`

This will store `⟨key⟩`'s value in `⟨macro⟩`. If `⟨pack-name⟩` or `⟨key⟩` doesn't exist, `⟨macro⟩` will be cleared.

`\PkgInfoDescription` `\PkgInfoDescription {⟨pack-name⟩}`

This will typeset a small paragraph (for data validation) with following `⟨pack-name⟩`'s info: `⟨name⟩`, `⟨version⟩`, `⟨date⟩` and `⟨description⟩`

For example:

```
\PkgInfoDescription{pkginfograb}
Package pkginfograb Version: 1.2 - 2025/11/16
Collecting package's info in a regular way
```

<code>\PkgInfoIfSet</code>	<code>\PkgInfoIfSet {<pack-name>} {<if-true>} {<if-false>}</code>
new: 2025/11/01	This will test if the given <pack-name> was set with <code>\PkgInfoSet</code> and <if-true> or <if-false> will be properly executed.

<code>\PkgInfoMapOver</code>	<code>\PkgInfoMapOver {<inline-code>}</code>
new: 2025/11/01	<inline-code> will receive (as <#1>) the name of each package set with <code>\PkgInfoSet</code> in the order they were set.

	Package pkginfograb Version: 1.2 - 2025/11/16 <i>Collecting package's info in a regular way</i>
<code>\PkgInfoMapOver</code>	Package codedescribe Version: 1.16b - 2025/10/30 <i>LaTeXCode Description/Documentation</i>
<code>{</code>	Package xpeekahead Version: 1.3a - 2025/11/01 <i>A simple peek ahead set up</i>
<code>\PkgInfoDescription{#1}</code>	Package codelisting Version: 1.16b - 2025/10/30 <i>LaTeXCode Listing</i>
<code>}</code>	